

# Cost-Volume-Profit Practice Problem

## Summit Recreation

Summit Recreation manufactures and sells a portable folding camp table. Management would like to analyze the company's current cost structure and profitability.

### Cost and Sales Information

The following information relates to the company's current year:

| Item                            | Per Unit |
|---------------------------------|----------|
| Selling price                   | \$120    |
| Direct materials                | \$32     |
| Direct labor                    | \$18     |
| Variable manufacturing overhead | \$10     |
| Variable selling expense        | \$8      |

| Additional Information                  | Amount       |
|-----------------------------------------|--------------|
| Annual unit sales                       | 12,000 units |
| Fixed manufacturing overhead            | \$300,000    |
| Fixed selling & administrative expenses | \$180,000    |

### Requirements

Using Excel, prepare a **contribution-margin income statement** and calculate the following:

1. Total sales revenue.
2. Total variable expenses.
3. Contribution margin.
4. Contribution margin per unit.
5. Contribution margin ratio (CM%).
6. Operating income.
7. Break-even point in units.
8. Break-even point in sales dollars.
9. Margin of safety in units, dollars, and percentage.
10. Degree of operating leverage.
11. Expected percentage change in operating income if sales volume increases by 10%.

12. Expected operating income if sales volume increases by 10%.

### **Target Profit Analysis**

Management would also like to earn **\$300,000 of operating income**. Determine:

13. The number of units that must be sold to earn the target operating income.

14. The amount of sales revenue required to earn the target operating income.